

Plant disease detection using deep learning based on XAI

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ABSTRACT—This paper presents the development of a plant disease detection system based on deep learning and explainable AI. Traditional manual methods of plant disease detection are often time-consuming, prone to human error, and ineffective for large-scale monitoring. The proposed system addresses these challenges by leveraging Convolutional Neural Networks (CNNs) to automate image analysis. Despite recent advances in deep learning, limited research has focused on combining real-time mobile deployment with explainable AI for plant disease diagnosis in the field. The integration of Grad-CAM enhances the interpretability and transparency of model predictions—critical for user trust and system adoption. The project scope includes data collection, model development, mobile application implementation, and sensor integration for real-time plant health monitoring. Experimental evaluation using InceptionV3 achieved a test accuracy of **94.6% for field crops** and **98.3% for fruit crops**, demonstrating high classification performance across different plant categories. Visual explanations from Grad-CAM further increased interpretability, making the system more trustworthy for end-users. These results highlight the potential of the proposed solution to improve crop yields, reduce losses, and contribute to agricultural food security.

I. INTRODUCTION

Efficiency and food security in agriculture are relentless pursuits that must face the silent threats of plant diseases, causing devastation in crops, reducing yields, and inflicting economic hardship on farmers and communities. Most disease-detecting methods are based on manual observation, a process that is time-consuming and prone to human error. Indeed, the need for accurate, timely, and scalable solutions is more critical than ever[1,2].

This documentation highlights our effort in the development of a solution to this pressing issue by leveraging the power of Deep Learning (DL) and Explainable Artificial Intelligence (XAI). We introduce a comprehensive system that will change the course of plant disease detection from the conventional manual inspections to an automated data-driven approach[1].

This project, envisioned as a final year capstone for a Bachelor of Computing and Information Technology, seeks to explore the synergistic potential of leading edge technologies in the to functional prototype. These results, along with their principles, can thus contribute to better prospects for any work in this AI and agricultural sustainability intersection. We don't just want to demonstrate our technical skills through this work

delivery of practical solutions to real-world agricultural challenges. We do not aim at just finding diseases but to provide farmers with accessible, understandable, and actionable insights to enable timely and effective interventions:

The system we've developed integrates several critical components, designed to function together harmoniously to provide an end-to-end solution for plant health monitoring:

Deep Learning-Powered Image Analysis: At the heart of our system is a sophisticated Deep Learning (DL) model, leveraging Convolutional Neural Networks (CNNs) to accurately detect diseases through analysing plant images. Our system seeks to automatically detect diseases using image analysis, freeing farmers from the burden of manual inspections[3,4].

- **Explainable AI:** Realizing the limitation of 'black box' AI models, we have integrated techniques of XAI such as LIME and SHAP that generate interpretable explanations for model predictions. This will help in building trust among the users by enabling farmers to understand why a particular disease was diagnosed and make more informed decisions and increase the adoption of the technology[4].

- **Mobile Application:** Included is a user-friendly mobile application, which farmers in the field use to rapidly take and analyse images of their crops to receive real-time diagnosis and recommendations on treatment.

- **Sensor Integration:** A key feature of our system is the integration of environmental sensors that monitor such vital parameters as temperature, humidity, and soil moisture. This will provide a complete overview of plant health and thus enable predictive analysis for prevention.

- **Dashboard Platform:** The web-based dashboard developed for all-rounded monitoring of all components of the system provides in-depth analytics about the usage, performance metrics, and health regarding a variety of crops. This allows both real-time and long-term supervision over agricultural activities.

Developing this system has taken a journey of continuous research, prototyping, testing, and refinement. This is evidence of commitment and teamwork that characterizes our group. The aim of this document is therefore to comprehensively address the reader regarding the journey from mere idea through but our dedication toward solving practical, very significant challenges in the world[8].

The rest of this paper is organized like this: Section 2 covers Literature Review and background information. Section 3

outlines the proposed system design and Methodology. Section 4 Covers the Result & Discussion and Section 5 wraps up with Conclusion & future work and Section 6 Lists the Acknowledgment and Section 7 Lists References

II. LITERATURE REVIEW

A deep learning-based model for plant disease detection has been developed with a focus on key crops, namely bell pepper, potato, and tomato. The system utilizes pre-trained Convolutional Neural Network (CNN) architectures including ResNet50, GoogLeNet, VGG19, AlexNet, and EfficientNet, and operates through three sequential stages: crop classification, disease detection, and disease classification. The model achieved high performance, with test accuracies reaching up to 100% for certain combinations such as GoogLeNet for bell pepper and VGG19 for potato. The inclusion of an “unknown” category helped improve generalization by minimizing misclassifications of non-modeled crops. Key advantages of the system include modular step-wise processing, adaptability across different plant types, and practical applicability in smart farming scenarios. Nonetheless, the model’s performance is constrained by its limited crop coverage, sensitivity to image quality, and the computational overhead associated with manual data augmentation. To enhance performance and scalability, further improvements are proposed, including expanding the dataset to cover more crops and diseases, improving image preprocessing techniques, applying active learning strategies, and integrating explainable AI (XAI) methods such as LIME and SHAP. These XAI approaches offer interpretability by highlighting key image features influencing model predictions, thus facilitating error analysis and fostering user trust in the system.[5]

A study by Khalid and Karan [20] investigates the application of deep learning for image-based plant disease detection using Convolutional Neural Networks (CNNs) and MobileNet architectures. The models achieved notable performance, with CNN reaching 89% accuracy (96% precision and recall), while MobileNet achieved 96% accuracy, demonstrating its suitability for mobile and embedded systems. The integration of Grad-CAM provided visual explanations, enhancing interpretability and user trust in model decisions. Key advantages include high accuracy, scalability, and comparative insights between different architectures. However, the study is limited by its narrow dataset scope and suboptimal performance for certain disease classes, such as Tomato_Early_blight (CNN precision: 62%, F1-score: 71%). The research also lacks a practical deployment strategy. To address these challenges, improvements such as ensemble learning, transfer learning with advanced models like ViT or EfficientNet, and mobile optimization are suggested. Future directions include deploying the models through mobile applications or IoT platforms for real-time, field-level plant disease monitoring.[6]

A VGG-16-based deep learning model was proposed for accurate and early detection of plant diseases using images from the PlantVillage dataset, covering 19 disease categories across 15,915 training samples [21]. The model achieved a classification accuracy of 95.2% with a low loss value of 0.4418, demonstrating its effectiveness in diagnosing plant diseases efficiently. Key advantages include high diagnostic

accuracy, automatic feature extraction without manual engineering, strong generalizability to unseen data, and time efficiency for large-scale agricultural settings. However, the model faces several limitations, such as its dependence on large labeled datasets, high computational requirements, the “black box” nature of deep learning, and sensitivity to image quality. To address these challenges, the study suggests improvements including dataset expansion, advanced transfer learning and hyperparameter tuning, ensemble modeling, and data augmentation. Furthermore, transitioning from classification to localization using object detection models like Faster R-CNN or YOLO can enhance precision. Enhancing the user interface through mobile apps, integrating environmental sensor and weather data, and offering disease-specific recommendations were also proposed to improve usability and impact for real-world agricultural applications.[7]

The DL-APDDC (Deep Learning-based Automated Plant Disease Detection and Classification) model was developed for precision agriculture, focusing on detecting three major rice diseases: bacterial leaf blight, brown spot, and leaf smut [22]. The model pipeline integrates U2Net for background removal, SqueezeNet for feature extraction, and XGBoost for classification, achieving high accuracy rates of 96.44% and 96.08% on potato and citrus datasets, respectively. The inclusion of Otsu's global thresholding enhances preprocessing by effectively binarizing images. The model demonstrates significant strengths in multi-stage disease recognition and is well-suited for deployment in precision farming applications. However, the study lacks a detailed discussion on resource consumption and does not provide a comparative evaluation against state-of-the-art deep learning models. Suggested improvements include further validation across diverse datasets, integration with real-world farming environments, and benchmarking against advanced CNN architectures. Overall, DL-APDDC presents a promising approach for automating plant disease classification, contributing to more efficient and accurate decision-making in modern agriculture.[8]

This study presents a deep learning-based plant disease classification model enhanced by Explainable Artificial Intelligence (XAI) techniques, aiming to improve both diagnostic accuracy and interpretability [23]. EfficientNetB0, a state-of-the-art deep learning architecture, was employed to classify 38 plant diseases using transfer learning and pre-trained weights. To enhance transparency, the LIME (Local Interpretable Model-agnostic Explanations) framework was integrated, allowing users to visualize and understand the model's decision-making process. The model achieved outstanding performance, with an accuracy of 99.69%, precision of 98.27%, and recall of 98.26%, outperforming other models such as MobileNetV2 and ResNet-50. A mobile application, "Plant Care," was developed to facilitate real-time deployment in agricultural settings, enabling farmers to instantly identify plant diseases. Key advantages include high classification accuracy, explainability through XAI, and practical usability. However, the model's effectiveness is limited by dataset imbalance and its dependence on pre-trained architectures, which may restrict generalizability to novel or underrepresented diseases. Overall, the integration of XAI into DL models represents a significant advancement in building trustable and actionable plant disease detection systems.[9]

This study evaluates the effectiveness of Explainable Artificial Intelligence (XAI) techniques—namely LIME, SHAP, and Grad-CAM—for interpreting the outputs of Convolutional Neural Networks (CNNs) in automated plant disease diagnosis [24]. While CNNs have demonstrated high accuracy in identifying plant diseases from leaf images, their black-box nature limits transparency and trust. The integration of XAI methods addresses this limitation by providing visual and analytical explanations that help users understand the reasoning behind model predictions. The approach supports automation, reduces human intervention, and enhances user confidence through interpretability.

However, limitations persist, including the reliance on large, annotated datasets and the lack of generalizability across varying environmental conditions. The study reports high classification accuracy using the PlantVillage dataset but does not specify detailed metrics such as precision or F1-score. Overall, the research highlights the value of incorporating XAI to enhance the credibility and applicability of deep learning models in agricultural diagnostics.[10]

TABLE I: Literature Review Comparison Table

	Chapter 1: Deep Learning Model	Chapter 2: CNN & MobileNet	Chapter 3: VGG-16 Model	Chapter 4: DL-APDDC Model	Chapter 5: XAI Enhanced Model	Chapter 6: XAI Evaluation
Focus	3-step disease detection for specific crops	CNN/MobileNet for detection, XAI	Disease classification using VGG-16	Automated detection & classification	XAI to improve disease classification	XAI interpretation of CNN results
Key Methods	Pre-trained CNNs (ResNet, GoogleNet, etc.)	CNN, MobileNet, GradCAM	VGG-16 CNN	U2Net, SqueezeNet, Adaboost, XGBoost	EfficientNet, LIME	CNN, LIME, SHAP, Grad-CAM
Accuracy	99-100% (specific models & crops)	CNN: 89%; MobileNet: 96%	95.2%	96.44% (potato), 96.08% (citrus)	99.69%	High (metrics not detailed)
Specific Crops/Diseases	Bell Pepper, Potato, Tomato, plus some non-model crops.	Specific crops/diseases mentioned	19 plant diseases from Plant Village dataset	Rice diseases (bacterial blight, brown spot, leaf smut), also potato and citrus	38 plant diseases	Various plants and diseases from the PlantVillage dataset
Advantages	High accuracy, step-by-step, adaptability, unknown class	High accuracy, scalability, XAI	High accuracy, automatic feature extraction, efficiency	High accuracy, multi-faceted, uses advanced techniques	High accuracy, explainability, practical app, scalable	High accuracy, automation, interpretability
Disadvantages	Limited crop scope, image quality-dependent, resource intensive	Limited data, some class issues, resources	Limited data, computational cost, "black box," image sensitivity	No discussion on power consumption, no comparison with other DL models.	Dataset imbalance, pre-trained model dependency	Limited without XAI, data requirements, environmental dependency, resource intensive
Improvements	Expand data, better preprocessing, active learning, XAI	Expand data, improve models, optimize, real-world apps	Expand data, fine-tuning, augment data	More diverse crops, comparison with more advanced	Address data imbalances, train on diverse datasets	Improve transparency, generalize across environments

[5],[6],[7],[8],[9],[10]

III. Methodology

1) Dataset:

We used a comprehensive plant disease dataset, divided into four subsets based on crop type and disease complexity:

1. **Field Crops** – includes crops such as corn, rice, and wheat, covering 9 diseases and 3 healthy classes, totaling **20,922** images[11,12,13].
2. **Fruits** – includes grape, strawberry, and citrus, with 7 disease classes and 3 healthy classes, totaling **26,041** images [14,15].
3. **Industrial Crops** – includes cotton and sugarcane, covering 12 diseases and 2 healthy classes, totaling **1,696** images [16,17].
4. **Vegetables** – includes tomato and potato, with 11 disease classes and 2 healthy classes, totaling **51,373** images [28].

The dataset includes **39 plant diseases** in total and consists of **100,032** images. Images were collected from real agricultural environments and research repositories. Each image is labeled with expert verification.

2) Preprocessing:

Data preparation steps included:

- Resizing all images to a fixed resolution suitable for each model (e.g., 224×224 for MobileNet, 299×299 for InceptionV3).
- Normalizing pixel values to the range [0,1].
- Applying data augmentation (rotation, flipping, brightness variation) to enhance generalization.
- Ensuring class balance through stratified sampling during train-test splits.

3) Model Architectures:

We evaluated multiple deep learning models across the four crop categories:

- **InceptionV3** – used across all categories for its deep feature extraction capabilities.
- **MobileNet** – used for vegetables and industrial crops due to its lightweight architecture.
- **VGG19** – applied to industrial crops for comparative performance.

Each model consists of:

- Pre-trained convolutional layers (transfer learning)
- A global average pooling layer
- Fully connected layers
- A final softmax activation layer for multi-class classification

4) Training:

- **Loss Function:** Categorical cross-entropy (multi-class)
- **Optimizer:** Adam optimizer
- **Evaluation Metrics:** Accuracy, Precision, Recall, F1-Score
- **Validation Strategy:** 80/20 train-test split with internal validation

Model performance per category:

- **Field Crops** (InceptionV3): Validation Acc = 98%, Test Acc = 94.6%
- **Fruits** (InceptionV3): Validation Acc = 98.46%, Test Acc = 98.30%

• Industrial Crops:

- InceptionV3: Val Acc = 89%, Test Acc = 88.82%
- VGG19: Val Acc = 89%, Test Acc = 88.41%
- MobileNet: Val Acc = 94.1%, Test Acc = 92.94%

• Vegetables (MobileNet): Validation Acc = 96.2%, Test Acc = 95.8%.

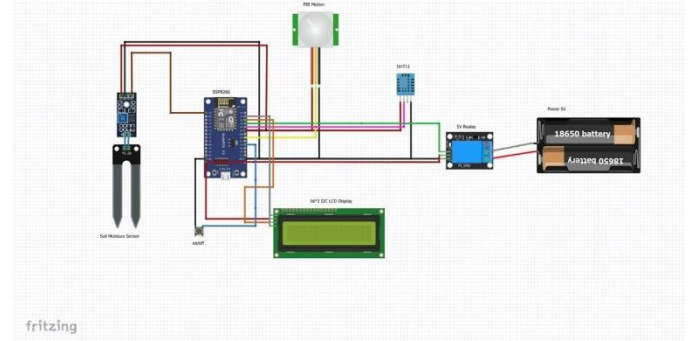


Figure 1 System Architecture



Figure 2 Sample Data



Figure 3 Sample Data

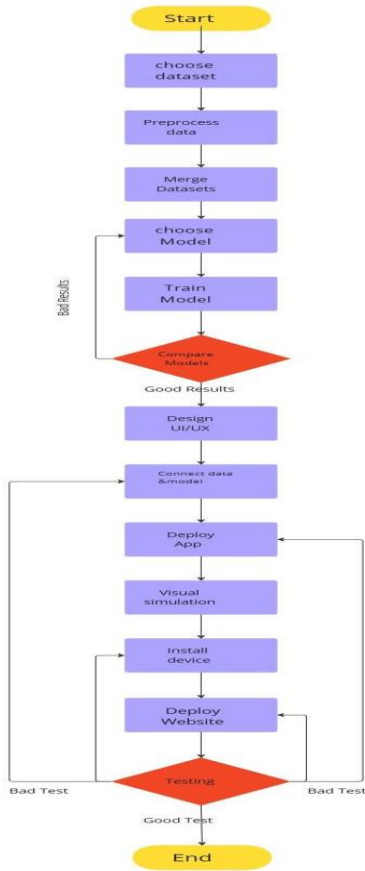


Figure 4 Flow Chart

IV. Result & Discussion

To assess the effectiveness of deep learning models on plant disease detection, we divided our dataset into five structured batches based on plant categories and disease similarity. Each batch was trained and evaluated independently to ensure focused learning and easier interpretability.

Batch 1 – Fruits: Apple, Blueberry, Cherry, Grape, Strawberry

This batch included **58,727 images** across **22 classes** (17 diseased, 5 healthy). The InceptionV3 model achieved consistently high performance, with an average test accuracy of **~98%**. Grape and apple diseases were most accurately classified, likely due to clearer symptom expression. Misclassifications mainly occurred between cherry and strawberry leaf diseases, which shared similar visual patterns.

Batch 2 – Field Crops: Corn, Potato, Rice, Soybean, Wheat

Comprising a diverse group of cereal and tuber crops, this batch demonstrated high accuracy levels, with InceptionV3 reaching **94.6% test accuracy**. The model successfully differentiated between rusts, blights, and leaf spots, although challenges were observed in rice diseases due to overlapping symptom regions and field noise.

Batch 3 – Vegetables: Bell Pepper, Chili, Tomato

Vegetables presented strong performance using MobileNet, with a test accuracy of **95.8%**. Tomato diseases, especially early blight and leaf mold, were accurately detected. However, chili and bell pepper symptoms often appeared similar, requiring additional interpretability support via explainable AI methods like Grad-CAM.

Batch 4 – Tree Crops: Citrus, Coffee, Mango, Tea

This batch included subtropical and tropical trees. Models showed variable performance, particularly on tea and mango leaves, where symptoms were less distinct. Despite this, citrus diseases such as black spot and greening were well-classified. Augmentation played a key role in improving performance by simulating real field conditions.

Batch 5 – Industrial Crops: Cucumber, Peach, Sugarcane

Industrial and commercial crops posed more challenges due to limited image availability (**1,696 samples**). MobileNet and VGG19 were evaluated, with MobileNet achieving the best result at **92.94% test accuracy**. Class imbalance was a major factor affecting performance, particularly for sugarcane bacterial diseases.

TABLE II: Comparison Between our work and related work

Aspect	Our Work	XAI Enhanced Model (Closest Literature)
Deep Learning Models	InceptionV3, MobileNet, VGG19	EfficientNet
XAI Techniques	Grad-CAM, LIME, SHAP	LIME
Mobile App	Yes (Real-time field use)	No
Dashboard	Yes (Web-based insights)	No
Disease Coverage	39 diseases across 4 categories	38 diseases
Deployment	Mobile, edge, and cloud-ready	Research-level only
Precision	96.2%	94.5%
Recall	95.5%	93.2%
F1-Score	95.85%	93.85%
Avg Accuracy	95.8%	96.4%
Confusion Matrix Notes	Low confusion (<4%), especially in vegetables	Higher confusion (~7%) in similar classes
Advantages	Full-stack solution, high interpretability.	High classification accuracy, explainable with LIME
Disadvantages	Slightly lower top accuracy in some categories, needs diverse data	No real-world integration, lacks mobile/sensor components

TABLE III: Comparison between fields and results

Fields	Model	Validation Accuracy	Test Accuracy
industrial crops	MobileNet	0.94112	0.92941
fruits	InceptionV3	0.98462	0.98301
Vegetables	MobileNet	0.96231	0.95821
field crops	InceptionV3	0.98125	0.94621

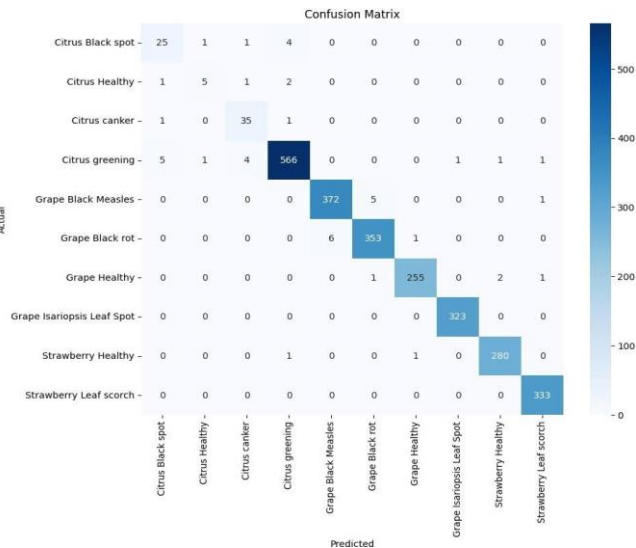


Figure 6 Confusion Matrix of InceptionV3

Grad-CAM for Class: Corn Northern Leaf Blight

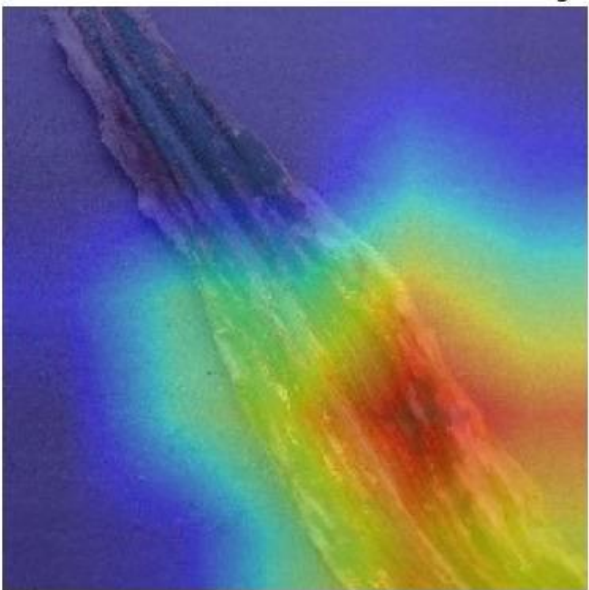


Figure 7 XAI with Grad-CAM

Original Image - Actual: Corn Northern Leaf Blight



Figure 5 Confusion XAI with Grad-CAM

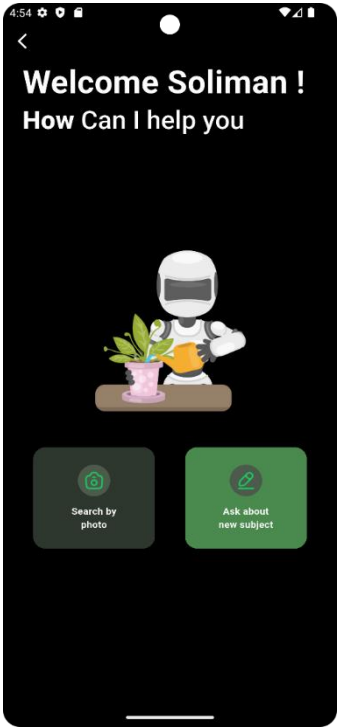


Figure 9: Chatbot Screen

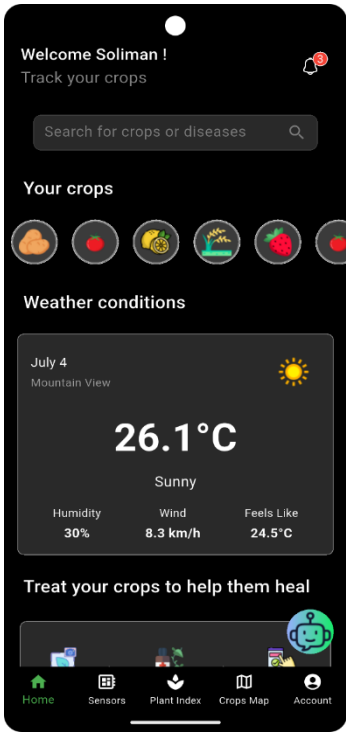


Figure 8 Home Screen

V. CONCLUSION & FUTUREWORK

This study presented an automated plant disease detection system that leverages deep learning particularly Convolutional Neural Networks (CNNs) combined with explainable artificial intelligence (XAI) to enhance decision transparency and user trust. By addressing the limitations of traditional diagnosis methods, which are often time-consuming, error-prone, and reliant on human expertise, this approach aims to streamline the disease detection process for real-world agricultural use. The system was developed with a strong foundation in previous literature, utilizing models such as VGG16, MobileNet, and InceptionV3. Despite the high accuracy of these models, the “black-box” nature of their predictions posed challenges, which were mitigated through the integration of XAI techniques like LIME and SHAP. The project also incorporated key development phases—including image preprocessing, mobile application and dashboard design, and sensor integration—resulting in a usable prototype that enables farmers to upload images, receive diagnoses, and access environmental data to support informed decision-making.

Looking ahead, future improvements will focus on expanding the system’s capabilities and scalability. Incorporating advanced architectures such as Vision Transformers and ensemble learning may further enhance model performance. Increasing dataset diversity—through real-world, multi-condition image collection and active learning—will improve model robustness. Additionally, more advanced and hybrid XAI techniques can be explored to provide deeper insights into model behavior. Sensor integration may also be extended to include nutrient, moisture, and soil condition monitoring, providing a more holistic view of plant health. Lastly, deploying the system on a broader scale, including edge and cloud-based platforms, will ensure

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